Learning and Assignment of IP address Manually to Computers.

Exp.No:03

Aim:

To understand the IP addressing schema and mannually add IP address to computer on network.

Theory:

1) An IP (Internet Protocol) address is an unique numerical label assigned to each device connected to a computer network.

2) IPv4 address are 32-bit numbers written in dotted-decimal notates.

3) IPv6 uses 128-bit address.

4) Subnet mask determines the network.

5) Manual (static) IP assignment is used for server and network devices requiring a permanent address.

6) DCP assigns IP dynamically.

Procedure:

1) In windows: open Network and Sharing Center -> Change adapter settings -> Right click Local Area connections -> Properties -> IPv4 -> Set IP manually.

2) Enter IP address : 192.168.1.10. Subnet Mask: 255.255.255.0, default gateway: 192.168.1.1.

3) Enter DNS Server: 8.8.8.8 (Google DNS). Click OK.

4) In linux! use command 'sudo ipaddress add 192.168.1.10/24 dev etho'.

5) To set gateway in linux 'sudo ip route add default via 192.168.1.1.'

6) for persistent configuration in Linux, edit /etc/network/interfaces or use nmcli.

7) verify the configuration using "ipconfig' or 'ip a'..

8) Ping the default gateway to configure network connectivity.

9) Ping another computer on the same subnet to verify communication.

Code:

```
sudo ip addr add 192.168..1.10/24 dev etho
sudo ip route add default via 192.168.1.1
ip a show etho
ping 192.168.1.1
nmcli con mod "wired connection1" ipv4.addresses
nmcli con mod 'wired connection 1' 192.168.1.10/24
                        ipv4.gateway.

nmcli con up 'wired connection1' mo
```

Result:

Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.